

Mp10 AutoTasks — Administrator's Guide

Backups, electronic remittances, modality worklist, and automated reports

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1 About this guide

Mp10 includes a background service called **AutoTasks**. It runs unattended on your server and takes care of work that would otherwise have to be done by hand every day:

- taking a **backup** of your data on a schedule you choose, archiving it and checking the archive is sound,
- **collecting files from your clearinghouse** — remittances, claim status responses and acknowledgements — and posting them into Mp10,
- reading **remittance PDFs** downloaded from insurer portals,
- asking insurers whether a patient's **coverage is active**,
- texting patients an **appointment reminder** they can reply to,
- keeping a **modality worklist** in step with the schedule, so imaging equipment can be told who its next patient is, and
- running **reports** on a schedule and e-mailing them to whoever needs them.

Each of those is turned on separately. Nothing in this list happens until you set the corresponding entry in the System Registry, with one exception noted in its own chapter: eligibility requests default to on.

This guide is for the person who administers Mp10 at the practice. It does not assume you are a programmer. It does assume you can open the **Admin10** module and that you know where your practice keeps its shared folders.

Anything in this guide that must be done by whoever installed Mp10 (your IT person or your reseller) is marked **IT task**.

A note on wording. Throughout this guide, *setting* means an entry in the Mp10 System Registry — not the Windows registry. They are unrelated. Nothing in this guide requires you to open Windows' own registry editor.

2 How AutoTasks works

AutoTasks runs as a Windows service on your Mp10 server. Once started it wakes up at a fixed interval, checks whether there is anything to do, does it, and goes back to sleep.

Setting	Section	Default	What it does
RUN_EVERY_N_SECONDS	AUTOTASKS	120	How often the service wakes up, in seconds

Two consequences worth understanding, because they explain most “why hasn’t it happened yet?” questions:

- **Nothing happens instantly.** If you drop a file into a folder, AutoTasks picks it up on its next wake-up — up to two minutes later by default.
- **Scheduled work fires on the first wake-up at or after the time you set**, not exactly on the second.

2.1 The log file

AutoTasks writes everything it does to a log file sitting next to the service program, named after it with a .out extension. On a typical install:

`C:\Mp10\AutoTasks.out`

This file is the single most useful thing to look at when something has not happened. It is plain text — open it with Notepad. Every line is stamped with the date and time.

The log is capped so it cannot fill the disk:

Setting	Section	Default	What it does
MAXIMUM_LOG_FILE_SIZE	GENERAL	8192	Largest log size in KB before it is trimmed

3 Where settings live: the System Registry

All three capabilities in this guide are switched on and configured in one place: **Admin10** → **Global System Registry**.

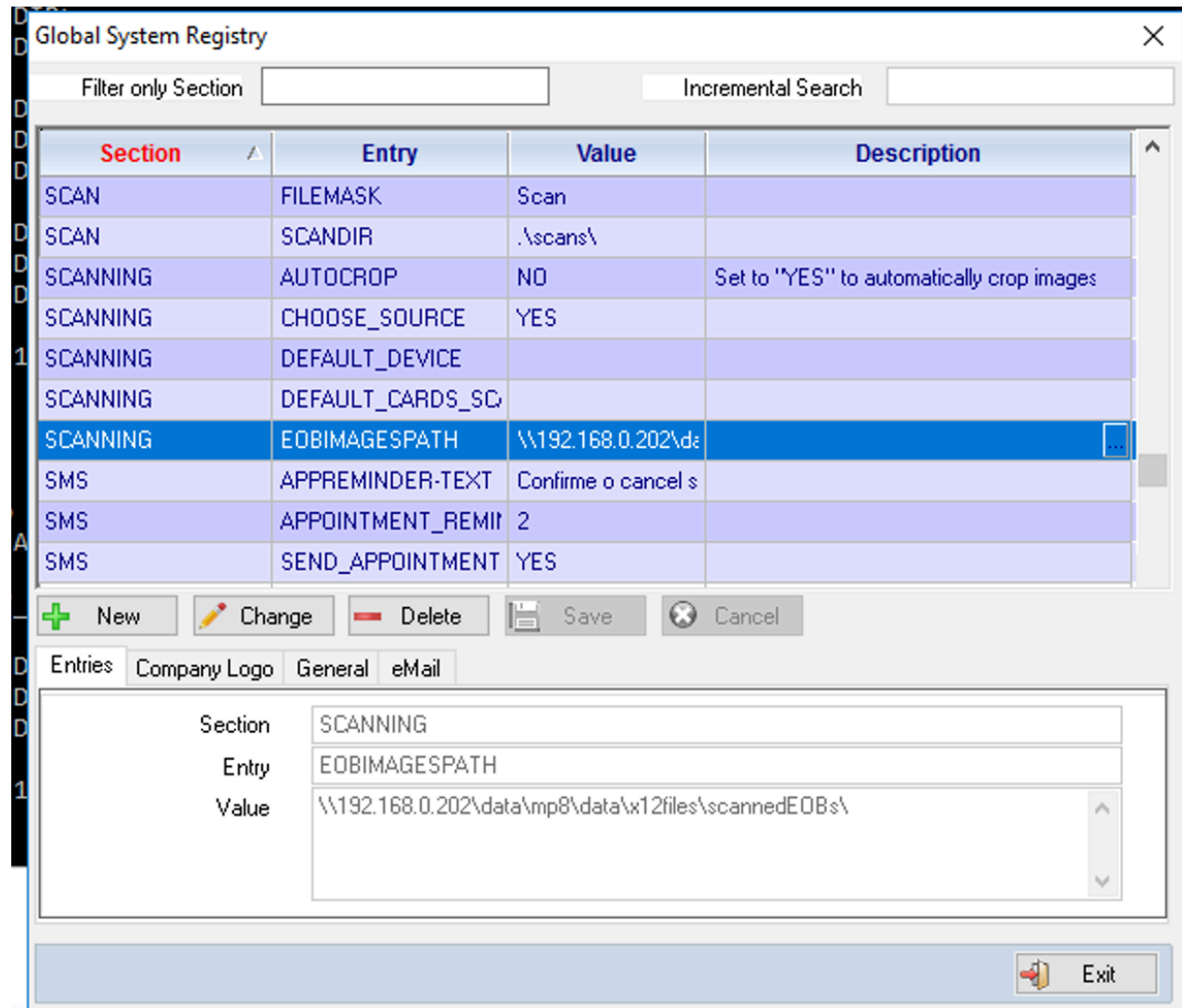


Figure 1: The Global System Registry in Admin10

Every setting has four parts:

Column	Meaning
Section	The group a setting belongs to, e.g. BACKUPS, SCANNING
Entry	The name of the setting, e.g. BACKUP_PATH
Value	What you are setting it to
Description	A free-text note for whoever reads it next

3.1 Adding or changing a setting

1. Open **Admin10** and go to the **Global System Registry**.

2. To find an existing setting, type into **Filter only Section** (for example BACKUPS) or use **Incremental Search**.
3. Press **New** to add a setting, or select a row and press **Change** to edit one.
4. Fill in **Section**, **Entry** and **Value** in the *Entries* tab at the bottom.
5. Press **Save**.

Type the Section and Entry names exactly as shown in this guide. They are matched by name. A misspelled entry is not an error — it is simply a setting nobody reads, so the feature stays switched off with no warning.

AutoTasks re-reads its settings at the start of every cycle, so a change you make in the System Registry takes effect within a couple of minutes. There is nothing to restart.

The exception is anything the service holds in memory rather than reads: the daily jobs remember that they have already run today, so a backup that has already failed today will not try again until tomorrow unless the service is restarted (see *Backups*).

4 Backups

AutoTasks can take an **Advantage Database Server hot backup** — a consistent copy of your live data taken while people are still working in Mp10 — and optionally compress it into a dated zip file, keeping only the most recent few.

4.1 Turning backups on

Backups run only when **all three** of these are set. Leave any one of them blank and no backup will ever run.

Entry	Example	What it means
BACKUP_PATH	\\adsserver\backup\mp10\	Where the backup image is written
BACKUP_TIMES	02:00:00	Earliest time of day the backup may start
BACKUP_DAYS	2345	Which days of the week it may run

All three go in section **BACKUPS**.

4.1.1 BACKUP_PATH — read this carefully

This is the one setting people most often get wrong.

The backup is written by the **Advantage Database Server**, not by Mp10 and not by your workstation. The path must therefore make sense *to the database server process*:

- **UNC paths work** — \\server\share\folder\
- **Local drive letters on the database server work** — D:\Backups\
- **Mapped network drives do NOT work** — Z:\Backups\ will fail, even if Z: is perfectly visible from your own PC

The account the Advantage service runs under must have permission to write there. (*IT task.*)

4.1.2 BACKUP_DAYS — the day codes

Days are given as a string of digits run together, one digit per allowed day:

Digit	Day
1	Sunday
2	Monday
3	Tuesday
4	Wednesday
5	Thursday
6	Friday
7	Saturday

So:

Value	Meaning
2345	Monday through Thursday

Value	Meaning
234567	Monday through Saturday
1234567	Every day
7	Saturdays only

4.1.3 BACKUP_TIMES

A 24-hour time, HH:MM:SS. 02:00:00 means “any time from 2 a.m. onward”.

Pick a time when the practice is quiet. A hot backup does not lock anybody out, but it does work the server hard.

A backup is attempted at most once per calendar day. Once AutoTasks has started one, it will not start another today, however many times it wakes up.

A failed attempt still counts as the day’s attempt. If the backup fails, AutoTasks does not keep retrying — it waits until tomorrow. This is deliberate: a hot backup runs for hours and hammering the database server all day cannot fix a wrong path or a full disk. It does mean **a failure needs somebody to act on it**, which is what the failure e-mail is for. If you fix the cause and want a backup the same day, restart the AutoTasks service.

4.2 Optional: zipping and retention

If you also set a zip path, AutoTasks compresses each backup into a dated archive, **reads that archive back to check it**, and only then prunes old ones.

Entry	Example	Default	What it means
BACKUP_ZIP_PATH	D:\Backups\Zips\	<i>(blank)</i>	Where dated zips are written. Blank = do not zip
MAX_BACKUPS_TO_KEEP	20	20	How many zips to keep. Oldest beyond this are deleted
FILES_TO_SKIP	oldtable.adt;temp.adt	<i>(blank)</i>	Data files to leave out of the backup, separated by semicolons

Zip files are named by date, so they sort chronologically:

Mp10Data-2026-07-31.zip

Unlike BACKUP_PATH, the zip path is used by AutoTasks itself, so an ordinary path reachable from the Mp10 server is fine.

MAX_BACKUPS_TO_KEEP deletes files. Once the number of zips in BACKUP_ZIP_PATH exceeds this number, the oldest are erased automatically. Keep that folder for backups only, and set the number to match the disk space you have.

4.2.1 Files that cannot be compressed are copied instead

Not all data compresses. A memo file holding scanned documents is already compressed — the scans are JPEGs or PDFs — and running it through compression again reads every byte for nothing, making the result very slightly *larger*.

AutoTasks samples each file before adding it, and simply copies the ones compression cannot help. The archive comes out the same size either way; the difference is that no time is wasted on files that were never going to shrink.

A practice storing images in Mp10 should not expect a small archive. If most of your data is scanned documents, the zip will be close to the size of the backup itself. That is the data, not a fault — and it means the archive folder needs roughly as much free space as the backup folder, for **each** archive you keep. Check `MAX_BACKUPS_TO_KEEP` against the free space you actually have.

4.2.2 The archive is checked before it counts as a backup

An archive is not treated as a backup until AutoTasks has opened it again and confirmed every file inside it is intact and the right size. Only then does it delete anything older.

This matters more than it sounds. If the archive cannot be written or does not check out, **nothing is deleted** — the previous archives are left exactly as they were, and you get a failure e-mail. Older backups are never traded away for a new one that might not work.

What happens to the failed archive depends on how it failed, because the two cases are worth different things to you.

It was written but did not check out. The archive is complete and wrong. That is worth keeping to look at, so it is renamed out of the way:

`Mp10Data-2026-07-31.failed`

Note the name replaces `.zip` rather than adding to it, so these can never be mistaken for archives or counted as one.

It ran out of room, or could not be written. The file is truncated. There is nothing to learn from half an archive, and keeping a partial copy of a very large backup on a disk that has just filled up only makes the next night worse — so it is deleted, and the failure e-mail says so.

Kept .failed files are pruned, but only the most recent few survive. They do not count towards `MAX_BACKUPS_TO_KEEP`; AutoTasks keeps the last three and erases older ones, naming each in the log and in the notice. Treat a `.failed` file as something to investigate, then delete it once you are done.

4.2.3 Restoring from an archive

To get files back out of an archive, use the archiving program directly. From the folder holding the AutoTasks program:

```
python backup_zip.py restore --zip "D:\Backups\Zips\Mp10Data-2026-07-31.zip" --target "D:\Restored"
```

To check an archive without extracting anything:

```
python backup_zip.py verify --zip "D:\Backups\Zips\Mp10Data-2026-07-31.zip" --mode full
```

`verify` finishes with a status of 0 when the archive is sound and 4 when it is not, and writes what it found to `backup_zip.result.json`. The `full` check reads every byte, so on a large archive it takes a while — that is why the nightly run uses a faster structural check instead, and why this thorough one is available to run by hand whenever you want reassurance.

Extracting files is only half of a recovery: what comes out of the zip is a backup *image*, not a database anyone can log into. Turning it back into one is covered in *Restoring the database* at the end of this chapter.

4.2.4 What must be installed for backups to work

What	Why	Where
Python 3	Creates and checks the zip archive	GENERAL / PYTHONPATH — folder holding <code>python.exe</code> , with a trailing backslash. Blank = rely on the system PATH
backup_zip.py	The archiving program itself	Must sit in the same folder as the AutoTasks program

If either is missing, no archive is created. The database backup itself still runs and your data is still written to BACKUP_PATH — but there will be no zip, and you will get a failure e-mail saying so.

4.3 Being told what happened

Backups run at night, when nobody is watching. AutoTasks e-mails a short report each time one finishes — **whether it worked or not** — so you find out without having to go and look.

Entry	Example	Default	What it means
NOTIFY_EMAIL	support@structuredsystems.com	support@structuredsystems.com	Who to notify. Blank = do not send

On a normal night you get **two** messages:

1. when the database backup finishes, and
2. when the zip archive finishes — this one also lists any older archives that were deleted to stay within MAX_BACKUPS_TO_KEEP.

The second message is the more useful of the two, because rotation is the only part of the process that destroys anything, and this is the only record you get of it.

The Mp10 backup archive was created and verified.

```
Data dictionary : \\adsserver\data\mp10\mp10.add
Archive         : D:\Backups\Zips\Mp10Data-2026-08-01.zip
Size            : 3072.0 MB (3,221,225,472 bytes)
Files archived  : 428
Source          : \\adsserver\backup\mp10\
Started         : 2026-08-01 04:37:11
Finished        : 2026-08-01 04:52:44
Duration        : 00:15:33
Verification    : headers - passed

Removed         : 2 older archive(s)
                  Mp10Data-2026-06-28.zip
                  Mp10Data-2026-06-29.zip
```

The **Verification** line is the one to read. If it is not there, or the subject says FAILED, nothing was deleted and the archive should not be relied on.

4.3.1 Sending to more than one person

NOTIFY_EMAIL takes as many addresses as you like. Separate them with commas or semicolons — you can mix the two, and spaces around them do not matter:

```
support@structuredsystems.com; manager@yourpractice.com, it@yourpractice.com
```

Anything that is not an address is ignored rather than sent, so one typo in the list does not stop the others receiving their copy. Look in `AutoTasks.out` for `Backup notice sent to` to see how many recipients it actually used.

Delivery uses the same `eMail` settings as automated reports (see *Making sure e-mail actually goes out*). If those are not configured, backups still run normally; only the notification is lost, and the log records that it could not be sent.

4.3.2 When something goes wrong

Failures are e-mailed too, with `FAILED` in the subject so you can spot or filter them. There are two, and they mean very different things.

Mp10 backup FAILED — the database backup did not happen. There is no new copy of your data today. The message carries the error the database server reported, which is usually enough to identify the cause:

The Mp10 hot backup FAILED. NO BACKUP WAS TAKEN TODAY.

```
Data dictionary : \\adsserver\data\mp10\mp10.add
Destination     : \\adsserver\backup\mp10\
Started        : 2026-08-01 02:00:03
Failed         : 2026-08-01 02:00:09
Ran for        : 00:00:06
```

```
ADS error       : 7077
ADS message     : The specified path is invalid
```

Mp10 backup archive FAILED — less serious. The backup itself worked and today's data is safe in `BACKUP_PATH`; only the zip step failed, so there is no dated archive for today and **nothing was rotated or deleted**. Usually a full disk, a permission problem on `BACKUP_ZIP_PATH`, or Python missing from the server.

The message names what actually went wrong and where to look. Two files next to the `AutoTasks` program carry the detail:

File	What is in it
<code>backup_zip.log</code>	What the archiver did, file by file, and why it stopped
<code>backup_zip.stdout.log</code>	Anything Python itself printed. Look here first if <code>backup_zip.log</code> does not exist at all — that combination means Python could not start

A failure e-mail means no backup today, and none is coming without you. `AutoTasks` does not retry after a failure — see *BACKUP_TIMES* above — so the message is not a warning about a passing glitch. Somebody has to fix the cause and, if the practice cannot go a day without a backup, restart the service.

Silence still means something is wrong: if the service itself is stopped, it cannot send anything at all. A night with no message of any kind is worth looking into.

4.3.3 E-mail has to be set up first

The notices go out through the practice's outgoing mail server, configured in the `eMail` section (see *Making sure e-mail actually goes out*). If that has not been set up, `AutoTasks` does not attempt to send at all — it writes a line to `AutoTasks.out` naming the settings that are blank and carries on.

Backups themselves are unaffected either way. Notification is the last step, and nothing about it can stop a backup from running or succeeding.

4.4 A working example — backups

Section	Entry	Value
BACKUPS	BACKUP_PATH	\\adsserver\backup\mp10\
BACKUPS	BACKUP_TIMES	02:00:00
BACKUPS	BACKUP_DAYS	234567
BACKUPS	BACKUP_ZIP_PATH	D:\Backups\Zips\
BACKUPS	MAX_BACKUPS_TO_KEEP	30
BACKUPS	NOTIFY_EMAIL	support@structuredsystems.com; manager@yourpractice.com
GENERAL	PYTHONPATH	C:\Program Files\Python313\

This takes a backup Monday through Saturday, any time after 2 a.m., zips it to a local disk, checks the archive, keeps the last 30 zips, and e-mails two people when it is done.

4.5 Backups are not a disaster recovery plan on their own

A backup that has never been restored is a hope, not a backup. Copy the zips off-site — or at minimum onto a different physical machine — and test a restore periodically. AutoTasks writes the file and checks that it is readable; making sure it survives a fire, a theft or a ransomware event is a decision only the practice can make.

Restoring is a single command, and it is worth running once so you know it works before you ever need it:

```
python backup_zip.py restore --zip "D:\Backups\Zips\Mp10Data-2026-07-31.zip" --target "D:\RestoreTest"
```

That command brings the files back out of the zip. Making a working database out of them is the next section.

4.6 Restoring the database

Unzipping an archive is not a restore. The files the nightly backup writes look like a database — tables, memo files, a data dictionary — but Advantage’s own documentation is blunt about them: they exist solely to be restored from, and reading them directly produces unpredictable results. Indexes are not in the backup at all. What turns a backup image back into a database people can log into is the **restore**, and it is done from the Admin program.

There are two steps. The first is only needed when the copy you want is inside a zip.

4.6.1 Step 1 — get the backup image onto a disk the server can reach

Last night’s backup is already sitting in `BACKUP_PATH`, unzipped. If that is the copy you want, skip straight to step 2.

For an older or off-site copy, extract the archive first, exactly as shown in *Restoring from an archive*:

```
python backup_zip.py restore --zip "D:\Backups\Zips\Mp10Data-2026-07-31.zip" --target "D:\RestoreTest"
```

Where you extract it matters. The restore in step 2 is carried out by the Advantage service **on the server**, so the same rule as `BACKUP_PATH` applies: the extracted folder must be on the server itself, or on a share the server machine can read. A drive letter that exists only on your workstation means nothing to the server.

4.6.2 Step 2 — restore with Admin10

Start Admin10 and log in with user code `adssys`. `ADSSYS` is Advantage’s own administrator account, not an ordinary Mp10 user — its password is the database administrator password, and the restore option does not appear on the menu for anyone else.

If your ADSSYS password is still whatever it was on installation day, change it. Anyone holding it can do everything described here, including overwriting a database.

Then pick **File → Restore DB From Backup**. It asks for three things:

Field	What to enter
Backup image .add path	The dictionary file inside the backup image — e.g. D:\RestoreTest\mp10.add, or the .add directly under BACKUP_PATH
Backup dictionary password	The ADSSYS password as it was when that backup was taken . If the password has changed since, an old image still wants the old password
Destination .add path	Where the restored database should be created — e.g. D:\RestoredDb\mp10.add

Both paths are read by the server, so the workstation-drive-letter rule above applies to them too. Both must end in .add.

The destination must never be the live data directory. A restore overwrites whatever is at the destination. Restore to a new or scratch directory, look at the result, and only then decide what to do with it — pointing the practice at a restored copy is a deliberate step, best taken with support on the line.

A restore can take a long time on a large database — every table is copied and every index is rebuilt from scratch, which is work the nightly backup skipped. A progress window shows that the restore is alive, how long it has been running and, once the server starts reporting it, how far along it is. Once the server starts reporting progress the restore can be cancelled from that window; a cancelled restore leaves an incomplete copy at the destination, which should be deleted before trying again.

When it finishes you have a complete, working copy of the practice’s database at the destination — dictionary, tables and freshly built indexes — completely separate from the live one. That is also the cheapest reassurance there is: restore into a scratch folder once in a while, log into the copy, and you know the night the archive is actually needed, it will work.

4.6.3 When there is no working database to log into

The instructions above assume you can start Admin10 normally. On the day you most need a restore that may not be true: Admin10 connects to a data dictionary at start-up, and the reason you are restoring may be that there is no longer a dictionary to connect to.

There is a way in, and it is worth knowing *before* the bad day.

Point Admin10 at the backup image itself. A backup image contains its own mp.add, so it is a data dictionary in its own right and Admin10 can log into it like any other. Start it with the backup folder as the path:

```
PrAdmin10.exe PATH:e:\adsdata\sfi-2026\backup\
```

Give the path of the *folder* holding the backup’s mp.add, with a trailing backslash — Admin10 adds the dictionary name itself. (The same thing can be done by changing the PATH entry in mp10.ini, but the command line leaves the file alone, which is one less thing to put back afterwards.)

Log in as adssys using **the ADSSYS password as it was when that backup was taken**, then use **File → Restore DB From Backup** exactly as described above. You are now connected to the backup, but you still restore *from* it, naming its .add as the source and a fresh folder as the destination.

Use that session for the restore and nothing else. Advantage is explicit that backup files exist to be restored from and that reading them directly “can produce unpredictable results”

— indexes are not even included in a backup. Admin10 may also tell you the data dictionary version is older than the program expects; that notice is informational and you should ignore it here. Never run a structure update against a backup image — that is your only copy of the data, and it is not the thing to be experimenting on.

If Admin10 cannot open the dictionary at all, it now offers the restore by itself. When the connection fails for any reason other than a rejected password, it asks:

```
The data dictionary could not be opened:  
  \\server\share\data\mp.add  
  ADS error 7077
```

Restore a database from a backup image now?

Answer yes and you get the restore dialog directly, without logging in — Advantage allows a restore to run on a bare connection to a directory, with no dictionary involved. Answer no and you get the familiar offer to create a new dictionary instead.

Two things about that prompt are deliberate. It does **not** appear when the password was simply wrong, so it is not a way around the login. And once the restore is done Admin10 closes, because it still has no dictionary — start it again pointing at the restored folder, and you are back to a normal session.

4.6.4 The restore utility: MpRestore10

Both routes above still go through Admin10, a program built around having a database. There is also a small separate one that is not:

MpRestore10.exe

It sits beside the other Mp10 programs and does exactly one thing. There is no login, because there is no dictionary to log into. It asks only for a folder on the server to connect through, and then shows the same restore dialog described earlier, with the same three fields.

The folder it asks for is **not** the database — it is any folder the Advantage service itself can read, used purely to reach the server. Your data folder is the obvious choice and is offered as a starting point (taken from mp10.ini), but if that is the part that has gone missing, any other folder on the server does just as well. The usual rule applies: it must be a path the *server* can resolve, so UNC paths and drive letters local to the server work, while a drive mapped on your own PC does not.

The Advantage service has to be running. A restore is carried out by the Advantage Database Server itself, not by the program in front of you. Advantage’s own documentation is flat about it: *“Online backup and restore functionality is only supported with the Advantage Database Server.”* Advantage Local Server — the mode where there is no service, just files on disk — cannot restore a backup at all, and no Mp10 program can work around that. If the machine holding your data has no Advantage service running, the service is the first thing to fix.

The path can also be given on the command line, handy for a recovery shortcut prepared in advance:

```
MpRestore10.exe PATH: \\server\share\mp10data\
```

Prepare this before you need it. Check once, on a calm afternoon, that MpRestore10 starts and connects — and while you are there, run a restore into a scratch folder with it. A recovery tool nobody has ever opened is the same kind of promise as a backup nobody has ever restored.

5 Automatic error report e-mails

When something goes wrong in a desktop program — PrBill10, Patients10, any of them — Mp10 already asks the person at the keyboard whether to send a report to Structured Systems. What this adds is what happens to that report **whether or not they say yes**: every error a desktop program hits is recorded regardless, and AutoTasks e-mails anything that has not already been sent on its next wake-up (every two minutes by default — see *How AutoTasks works*, above). A user declining to send no longer means the report is lost; it just waits for AutoTasks to pick it up.

5.1 Errors are grouped, not listed one by one

The same error often happens more than once — on the same workstation, or on several. AutoTasks does not send one e-mail per occurrence. It sends **one message per distinct error**, however many times it happened, and the message says how many times and where.

Each message tells you:

- which program and which build,
- which workstations and users hit it, and when it was first and last seen, and
- the list of program locations the error passed through — the stack — so Structured Systems can see where it happened.

It deliberately leaves out the detailed diagnostic dump — the full parameter and variable listing stays in the database, not in your inbox. **It also leaves out machine details**: CPU, memory and the free-resources figures are recorded with the error but never e-mailed. If Structured Systems needs those to chase a resource-exhaustion problem, they come from the database, not from this message.

5.2 Turning it on

It ships switched off. There is no default address — until an administrator sets one, nothing is sent, ever.

Entry	Example	Default	What it means
NOTIFY_EMAIL	support@structuredsystems.com	no one — blank	Who to notify. Blank = do not send. This is what switches the feature on
MAX_ROWS_TO_KEEP	1000	1000	How many error reports to keep before the oldest already-sent ones are deleted. Pruning is part of the error-report pass, so while NOTIFY_EMAIL is blank, or while e-mail itself is not set up, this setting does nothing at all . Once both are in place it is applied on every wake-up, whether or not there is anything to send — so lowering it takes effect within a couple of minutes on a quiet site, not only after the next error

Entry	Example	Default	What it means
MAX_GROUPS_PER_RUN	10	10	Most distinct errors e-mailed on any one wake-up; anything left over waits for the next one

All three live in section **ERRORREPORTS**.

Delivery uses the same mail settings as backups and automated reports — see *E-mail has to be set up first*, above. If those are not configured, AutoTasks does not attempt to send an error report either, and the log records that it could not be sent.

5.3 A worked example

This is reproduced exactly as AutoTasks composes it — labels, spacing and all — because that is what will actually land in your inbox:

Subject: [PrBill10.exe] BASE/1005|POSTPAYMENT|2214 - 4x on 2 station(s)

Signature : BASE/1005|POSTPAYMENT|2214

Program : PrBill10.exe build 931

Occurred : 4 time(s)

First seen: 2026-07-30 09:14:02

Last seen : 2026-08-01 11:47:55

Stations and users (oldest to newest)

FRONTDESK-2	jsmith	2026-07-30 09:14:02	ErrorNum 8841
FRONTDESK-2	jsmith	2026-07-30 15:02:11	ErrorNum 8852
BILLING-1	mriviera	2026-08-01 08:30:47	ErrorNum 8901
BILLING-1	mriviera	2026-08-01 11:47:55	ErrorNum 8919

Stack (most recent report)

Called from:POSTPAYMENT(2214)

Called from:APPLYBALANCE(861)

Called from:_BEND(0)

Full detail -- including the variables dump and machine/resource stats omitted from this e-mail -- is in

The program name in both the subject and the **Program** line is the file name of the running executable, spelled exactly as it sits on disk — **PrBill10.exe**, not **PRBILL10.EXE**. The **build** figure is the internal build number the program was compiled with (a plain counter such as 931), not the dictionary version.

The signature — **BASE/1005|POSTPAYMENT|2214** above — is an internal fingerprint, not a sentence. It is the part that decides which occurrences belong together: the error subsystem and code (**BASE/1005**), then the routine and the line number inside it. Two occurrences with the same signature are treated as the same error and arrive in one message.

The first line of the message and its closing sentence use internal labels rather than plain English. They are quoted here exactly because that is what will arrive in your inbox — the rest of this guide talks about “the error report” and “the record kept in the database,” not those names.

5.4 If sending fails

If the mail server is unreachable or rejects the message, AutoTasks retries on later wake-ups. After three failed attempts it gives up on that report and records the fact in the log — see *The log file*, above.

5.5 What gets deleted, and what never does

Once the number of error reports on file goes over `MAX_ROWS_TO_KEEP`, AutoTasks deletes the oldest ones to bring it back down — but only reports that have already been sent. **A report that has never been sent is never deleted to make room for a new one.** If your site is generating errors faster than they can be e-mailed, the table grows past the keep-limit rather than losing a report nobody has seen yet.

5.6 Where to look when nothing arrives

Everything this feature does — including whether each message actually got out — is written to `AutoTasks.out`, the same log backup notices use. If an error happened and no e-mail followed, that is the first place to look.

6 Electronic remittances: the two kinds

Payment advices reach a practice in two different shapes, and AutoTasks handles them with two separate mechanisms. They are configured independently, in different sections of the System Registry, and turning one on does nothing for the other.

	X12 835 files	EOB / ERA PDFs
What it is	The standard electronic remittance format, a data file	A printed advice saved as a PDF
Where it comes from	A clearinghouse, usually downloaded automatically	Downloaded by hand from an insurer's portal
Turned on by	AUTOTASKS / PROCESS_835	SCANNING / EOBIMAGESPATH
Folder settings live in	X12FILE_PATH	SCANNING
Needs Python	No	Yes

If your clearinghouse delivers 835 files, that is the path to use — it carries more detail and needs no interpretation. The PDF route exists for payers who will only give you a printed advice.

Both post into the same remittances table once processed.

7 Collecting files from the clearinghouse

Rather than have somebody log into the clearinghouse portal and download files by hand, AutoTasks can collect them itself on a schedule. One mechanism brings down everything the clearinghouse has routed to you.

7.1 What gets collected

The clearinghouse tags each file it sends with a document type. AutoTasks saves each one with that type at the front of the file name, so you can tell at a glance what arrived:

Arrives as	What it is	Processed by
835_....txt	An electronic remittance — a payment	<i>X12 835 electronic remittances</i>
277_....txt	A claim status response — what the payer did with a claim	<i>Claim status responses (277)</i>
EDI_....txt	An acknowledgement or notice about a claim submission	<i>Claim acknowledgements (EDI)</i>

Everything lands in one folder — the one named by `X12FILE_PATH` / 835 — whatever its type. That folder is the clearinghouse in-tray, not the “835 folder”, despite the setting’s name. Each processor then picks out the files belonging to it. This is why the 835 processor only looks at names containing 835.

A file that has already been collected is not written a second time: AutoTasks checks both the in-tray and the history folder before saving.

7.2 Turning collection on

Section	Entry	Example	What it means
AUTOTASKS	<code>GetRoutedFiles</code>	YES	Collect files from the clearinghouse. Anything other than YES = off
AUTOTASKS	<code>GetRoutedFilesStartTime</code>	07:00:00	Earliest time of day it may run. Blank = no lower bound
AUTOTASKS	<code>GetRoutedFilesEndTime</code>	19:00:00	Latest time of day it may run. Blank = no upper bound
AUTOTASKS	<code>GetRoutedFilesFrequency</code>	60	Minutes to wait between runs on the same day
AUTOTASKS	<code>GetRoutedFilesDays</code>	23456	Which days it may run on, using the same day codes as <code>BACKUP_DAYS</code>

Collection stops at 7 p.m. regardless of what you set. There is a fixed cut-off in the program, described in the log as “*Past working hrs not fetching routed files to give Inmediata a rest.*” Setting `GetRoutedFilesEndTime` later than 19:00:00 will not extend collection past it.

7.3 The clearinghouse account

Section	Entry	What it means
X12WebService	UserName	Clearinghouse user name
X12WebService	Password	Clearinghouse password
X12WebService	FileTransfer_Url	The file-transfer service address
X12WebService	TimeOut	How long to wait for a response, in milliseconds. Default 30000
X12WebService	MarkDownloadsAsDownloaded	NO (recommended) — see below

7.4 MarkDownloadsAsDownloaded — leave this alone unless asked

Once a file is marked as downloaded, the clearinghouse will not offer it again. When that marking happens is the whole question.

- **NO (the default, and the safe choice).** AutoTasks fetches the files, writes them to disk, and *then* tells the clearinghouse they were collected. If the download is interrupted or the disk is full, nothing was marked, so the clearinghouse still has the files and the next run collects them again.
- **YES.** The files are marked as downloaded as part of the fetch itself, before AutoTasks has written anything. If the run fails at that point, the clearinghouse considers the files delivered and **will not offer them again** — they would have to be retrieved from the portal by hand.

Set it to YES only if the clearinghouse specifically asks you to.

7.5 Checking it is working

```
GetRoutedFiles
```

```
7 routed files downloaded
```

```
\\server\data\x12files\835_20260803074113.txt
```

```
\\server\data\x12files\277_20260803074115.txt
```

A run that finds nothing logs `0 routed files downloaded`, which is normal outside business hours or when the clearinghouse has nothing new.

8 X12 835 electronic remittances

An 835 is the standard electronic remittance file. AutoTasks can collect them from the clearinghouse, read them, post the payments, and file the used ones away — with no manual step at all once it is set up.

8.1 The journey of an 835 file

```
clearinghouse
|
| (1) downloaded -- AUTOTASKS / GetRoutedFiles
v
X12FILE_PATH / 835          <- the working folder
|
| (2) processed -- AUTOTASKS / PROCESS_835
v
payments posted into Mp10
|
| (3) filed away
v
X12FILE_PATH / 835_COPY_PATH <- the history folder
```

The file is **copied** to the history folder and then **erased** from the working folder. Erasing is the point: it is what stops the same remittance being processed again the next time round.

8.2 Step 1 — Create the two folders

You need two folders, and **both must exist before you start**. AutoTasks does not create them for you.

Folder	Purpose
Working folder	Where 835 files arrive and wait to be processed
History folder	Where they are filed after they have been posted

Requirements:

- The AutoTasks service account needs permission to **read, write and delete** in the working folder — it erases files there once they are safely filed.
- It needs permission to **write** in the history folder.
- Keep them separate. Do not make the history folder a subfolder that the working folder's file search would find.
- Do not use the working folder for anything else.

8.3 Step 2 — Declare the folders

Section	Entry	Example value	What it means
X12FILE_PATH	835	\\server\data\x12files\The working folder — where 835 files are read from	
X12FILE_PATH	835_COPY_PATH	\\server\data\x12files\History folder — where they are moved after posting	

Include the trailing backslash on both.

835_COPY_PATH is not optional in practice. If you leave it blank, processed files are never erased from the working folder, so AutoTasks finds and reprocesses the same remittances on every single cycle, for ever. The same happens if the history folder does not exist or cannot be written to: the copy fails, so the erase is skipped. **If you see the same remittances being processed over and over, this is why.**

8.4 Step 3 — Turn processing on

Section	Entry	Value	What it means
AUTOTASKS	PROCESS_835	YES	Read and post 835 files. Anything other than YES = off

8.5 Step 4 — Optional: collect the files automatically

AutoTasks can fetch files from the clearinghouse itself, so nobody has to download anything by hand. That mechanism collects remittances, claim status responses and acknowledgements together — see *Collecting files from the clearinghouse* below.

If you leave `GetRoutedFiles` off, everything else still works; somebody just has to put the 835 files into the working folder themselves.

8.6 Two things that catch people out

The file name has to contain 835. AutoTasks only looks at files whose name includes those three digits — `*835*`. A remittance saved as `remit_0731.txt` will sit in the folder untouched, for ever, with nothing in the log to say why. If files are being ignored, check the name first.

Files older than six months are skipped. Anything more than 180 days old is left alone and noted in the log as `File more than 6 mo old`. This stops an old archive folder being reposted by accident. If you genuinely need to post something older, it has to be done by hand.

8.7 A working example — 835 files

Section	Entry	Value
AUTOTASKS	PROCESS_835	YES
X12FILE_PATH	835	\\server\data\x12files\
X12FILE_PATH	835_COPY_PATH	\\server\data\x12files\history\
AUTOTASKS	GetRoutedFiles	YES
AUTOTASKS	GetRoutedFilesStartTime	07:00:00
AUTOTASKS	GetRoutedFilesEndTime	19:00:00
AUTOTASKS	GetRoutedFilesFrequency	60
AUTOTASKS	GetRoutedFilesDays	23456

This collects remittances from the clearinghouse once an hour, Monday to Friday, between 7 a.m. and 7 p.m., posts them, and files them into the history folder.

8.8 Checking it is working

At startup AutoTasks writes the state of each X12 folder into `AutoTasks.out`:

```
X12 835 dir  \\server\data\x12files\  exists
```


If that line says the folder is missing, nothing else in this chapter will work. During a run you will see how many files were found:

```
ProcessAll8350nDir 12 Files found on \\server\data\x12files\
```

Then, as each is filed away, the copy and the erase are both logged. A file that is found but never erased is the signature of a history-folder problem — see the warning under *Step 2*.

9 Claim status responses (277)

A 277 tells you what the payer did with a claim you submitted — accepted, rejected, pending, and why. AutoTasks reads them and posts the status against the claim, so the billing staff can see it without logging into a portal.

Section	Entry	Value	What it means
AUTOTASKS	PROCESS_277	YES	Read and post 277 files. Anything other than YES = off
X12FILE_PATH	277	\\server\data\x12files\	The folder 277 files are read from
X12FILE_PATH	277_COPY_PATH	\\server\data\x12files\history\	Where they are filed after posting

The same rule as for 835 files applies: **the history folder must exist and be writable**, or the file is never erased and gets reprocessed on every cycle.

X12FILE_PATH / 277 and X12FILE_PATH / 835 should normally name the same folder.

Automatic collection puts *everything* it downloads into the folder named by the 835 entry — including 277s. If the 277 entry points somewhere else, downloaded 277 files will sit in the in-tray and never be processed. Unless you have a specific reason to separate them, set both to the same path.

AutoTasks accepts three shapes of status file: a standard X12 277, an XML status report, and a tab-delimited validation report. It works out which it is by reading the file, so you do not have to tell it.

A file that is not any of those is left alone and logged as **Unlegit 277**. That line is normal in a shared in-tray — it is what an 835 looks like to the 277 reader — and is not an error.

10 Claim acknowledgements (EDI)

When you submit claims, the clearinghouse sends back acknowledgements saying it received them. AutoTasks reads these and records them against the claims.

Section	Entry	Value	What it means
AUTOTASKS	ProcessEDIs	YES	Read acknowledgement files. Anything other than YES = off
X12FILE_PATH	277	\\server\data\x12files\	Read from the same folder as 277s
X12FILE_PATH	277_COPY_PATH	\\server\data\x12files\History\	Files are afterwards

There are no separate path settings — acknowledgements use the 277 pair.

Low-level acknowledgements are deleted, not filed. Files named TA1_..., _TA1_..., 999_... and _999_... are transport-level receipts that carry nothing a practice needs. AutoTasks erases them on sight to stop them accumulating. They are **not** copied to the history folder first, so if you want to keep them for an audit, this is not the mechanism that will do it.

11 Insurance eligibility (270 / 271)

AutoTasks can ask insurers whether a patient's coverage is active — a 270 request out, a 271 response back — and store the answer against the patient, so front-desk staff see current coverage without phoning anyone.

Section	Entry	Default	What it means
AUTOTASKS	AutoSend_270	NO	Send eligibility requests. Anything other than YES = off
X12FILE_PATH	270	\\server\data\x12files\	Working folder for eligibility files

On an existing installation this is probably already set to YES. Earlier versions turned eligibility on by default, and that value was written into the System Registry the first time the service ran. Changing the default does not change what is already stored. If you do not want eligibility requests going out, check the `AutoSend_270` entry and set it to `NO` yourself.

11.1 When it happens

There is no schedule to configure. Eligibility runs on **every service cycle**, alongside the other tasks, and asks about three groups:

1. **Patients** whose insurance record was added or changed recently
2. **Claims** awaiting submission
3. **Encounters** being prepared

11.2 Why a patient might not be checked

This is the part that generates support calls, because nothing appears in the log to explain a patient who was skipped. A patient's insurance record is only asked about when **all** of these are true:

- the insurance plan is flagged to fetch eligibility — `isFetch271` ticked on the plan record in **Planfile**
- the plan has its **Submitter**, **Receiver** and **GS03** identifiers filled in
- the insurance record was added or changed **within the last 7 days**
- there is a **member/contract number** on the record
- it is the patient's **primary** insurance
- no eligibility response has already come back since the record was last changed

The first two are the usual culprits, and they live on the **plan**, not the patient. If eligibility works for one insurer and not another, compare their plan records before looking anywhere else.

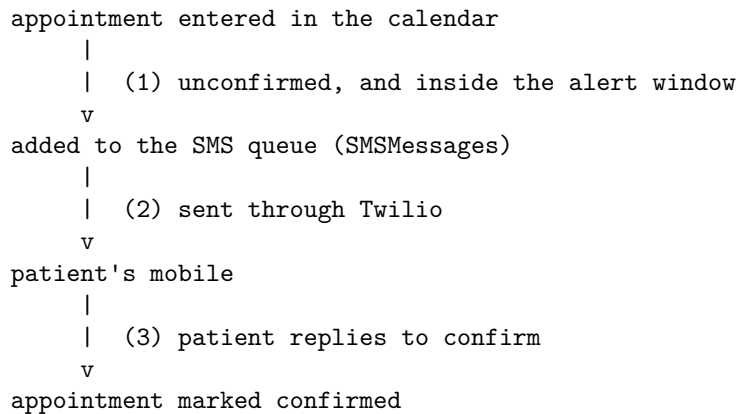
The 7-day window is deliberate: it checks coverage that has recently changed rather than re-asking about every patient in the practice every day. To force a re-check on an older record, open and re-save it — that updates its modified date and brings it back into the window.

12 Appointment reminders by text message

Appointments entered in the calendar can be confirmed by text, so the practice finds out about a cancellation before the slot is wasted rather than after.

AutoTasks watches **every appointments table in the data dictionary** — any table whose name contains `appntmnts` — so a practice running more than one calendar is covered without extra configuration.

12.1 How a reminder happens



Queueing and sending are separate steps, both on every service cycle. A message that cannot be sent stays in the queue and is retried.

12.2 The settings

Section	Entry	Example	Default	What it means
SMS	SEND_APPOINTMENT_REMINDERS	RESENDERS	NO	The on/off switch
SMS	APPOINTMENT_REMINDERS_ALERT_DAYS	RESENDERS	3	How many days ahead to look. Must be more than 0 or nothing is sent
SMS	APPREMINDER-TEXT	see below	<i>(blank)</i>	The message itself
SMS	COOLING_PERIOD	1	1	Hours to wait after an appointment is changed before texting about it
SMS	MAX-DAILY-SMS	100	100	Most messages to send in one day, as a spend guard
SMS	SMS_STATUS_CALLBACK(<i>URL address</i>)	URL	<i>(blank)</i>	Where Twilio reports delivery status. Blank = no delivery reporting

The Twilio account lives in its own section — **IT task**:

Section	Entry	What it means
TWILIO	ACCOUNT_SID	Twilio account identifier
TWILIO	AUTH_TOKEN	Twilio authentication token
TWILIO	TEL_FROM	The number messages are sent from

12.3 Writing the message

APPREMINDER-TEXT is the text the patient receives. Three placeholders are replaced as each message is built:

Placeholder	Replaced with
[date]	The appointment date and time, e.g. 08/05/2026 2:30 PM
[tel]	The patient's mobile number
[guid]	The appointment's internal reference, for use in a confirmation link

For example:

Reminder: your appointment is on [date]. Reply C to confirm or R to reschedule.

Keep it short. Long messages are split and billed as more than one.

12.4 Which appointments get a reminder

An appointment is texted when **all** of these are true:

- it is in the future, and within APPOINTMENT_REMINDERS_ALERT_DAYS days
- it has **not** already been confirmed or cancelled
- no reminder has been sent for it already
- there is a mobile number of at least 10 digits on the appointment
- the appointment has a real time on it, not 00:00:00
- it was last changed more than COOLING_PERIOD hours ago

One text per patient per day, however many appointments they have. If a patient is booked twice on the same day, they get a single reminder about the first. This is deliberate — it stops a patient with a morning and an afternoon slot receiving two texts.

12.5 Keeping the cost under control

MAX-DAILY-SMS is a hard stop. Once that many messages have been sent today, the queue is left alone until tomorrow and the log records why:

Maximum number of SMS reached. Total used: 100 Maximum set to: 100

If reminders stop partway through a busy day, this is the first thing to check. The messages are not lost — they stay queued and go out the following day, by which time the appointment may have passed.

13 Electronic remittances (EOB / ERA PDFs)

Insurers publish payment advices — remittance notices, ERAs, EOBs — to their provider portals as PDF files. AutoTasks can read those PDFs and post the payments into Mp10 automatically, so nobody has to key them in.

This is the **PDF** route. If your payer sends X12 835 files instead, see the previous chapter — the two are configured separately.

13.1 What you have to do

1. **Choose a folder** where remittance PDFs will be saved.
2. **Declare that folder** in the System Registry.
3. **Save PDFs into it** as you download them.

That is the whole workflow. Everything after step 3 is automatic.

13.2 Step 1 — Choose the folder

Any folder the Mp10 server can reach. A share on your file server is the usual choice, so that billing staff can drop files in from their own workstations.

Requirements:

- The AutoTasks service account needs permission to **read, write and delete** in it. It does not merely read the PDFs — it moves them when it is done.
- If more than one person will save files there, give them all write access.
- Do not point it at a folder used for anything else. AutoTasks processes and then moves *every* PDF it finds there.

13.3 Step 2 — Declare the folder

Section	Entry	Example value
SCANNING	EOBIMAGESPATH	\\192.168.0.202\data\mp8\data\x12files\sca

The screenshot in *Where settings live* shows exactly this entry being edited.

Setting this value is the on/off switch. There is no separate “enable” setting. If EOBIMAGESPATH has a value, remittance processing is on; if it is blank, it is off. Include the trailing backslash.

It takes effect on the next cycle — a couple of minutes — with nothing to restart.

13.4 Step 3 — Save your remittance PDFs there

Download the remittance PDF from your insurer’s portal as you normally would, and save it into the folder. Nothing else is required — you do not need to rename the file or record it anywhere.

Guidelines:

- **One remittance per PDF.** If the portal offers a combined download of many checks, prefer separate files.
- **Let the download finish before AutoTasks wakes up.** Saving directly into the watched folder over a slow connection can expose a half-written file. If that is a risk, save to your desktop and move the finished file in.
- **Do not put anything else in the folder.** Non-remittance PDFs will be read, found to contain nothing usable, and left behind.

13.5 What happens next

On its next wake-up, for each PDF in the folder, AutoTasks:

1. **Reads the PDF.** It first tries to read the text directly. If the PDF is a scan with no text in it, it falls back to optical character recognition.
2. **Writes a ..._eob.json file** beside the PDF holding what it found — the check number and date, the payee, and every claim and service line.
3. **Checks the remittance belongs to your practice** (see below).
4. **Posts the payments** into the remittances file, ready to be applied.
5. **Moves the PDF** into a **history** sub-folder so it is not processed twice:

...\scannedEOBs\history\C6472004.pdf

Create the history sub-folder yourself. Unlike the 835 history folder, this one has no setting — it is always a folder called **history** directly inside **EOBIMAGESPATH**, and AutoTasks does not create it. If it is missing, or the service account cannot write to it, the PDF cannot be moved and the same remittance is read again on every cycle.

If you want to know what happened to a particular file, search **AutoTasks.out** for its name.

13.6 If your server serves more than one practice

A single AutoTasks server can serve several practices, each with its own data dictionary, all watching the same folder.

AutoTasks reads the **payee** — the provider the cheque was made out to — from each remittance and compares it against the NPIs and provider numbers registered for the practice it is currently connected to.

- If they **match**, the remittance is posted and the PDF is moved to **history**.
- If they **do not match**, the file is **left exactly where it is**, untouched, so the run connected to the practice that owns it can pick it up.

You do not need to sort remittances by practice, and nothing is lost if a file is downloaded into the wrong practice's queue.

If a remittance carries no identifiable payee at all, it is treated as yours and processed. Most payer layouts do print one; some do not.

13.7 What must be installed on the server

(IT task — but if remittances are not being read, this is the first thing to check.)

Component	Purpose	Registry setting
Python 3	Runs the extractor	GENERAL / PYTHONPATH — folder holding python.exe , with a trailing backslash
extract_eob.py	The extractor itself	Must sit in the same folder as the AutoTasks program
Poppler	Renders scanned PDFs to images	Installed at C:\Mp10\poppler\Library\bin
Tesseract OCR	Reads text out of scans	Installed at C:\Program Files\Tesseract-OCR

Poppler and Tesseract are needed **only for scanned remittances**. If your insurers all publish normal text PDFs, remittances will process without them.

A second log, `python_debug.log`, is written next to the AutoTasks program and holds the extractor's own output. It is the place to look if a specific PDF fails.

14 Automated reports

Any Mp10 report can be scheduled to run by itself and be e-mailed out. You set this up from the report itself, not from a central screen.

14.1 Setting up a report

Open the report you want as you normally would, set its parameters, then go to the **Automate Report** tab.

The screenshot shows the 'Unbilled Report Parameters' dialog box with the 'Automate Report' tab selected. The 'Automated Report Users' table lists 'RCB' with email 'reinaldo.crespo@gmail.com'. The 'Frequency' section has 'Daily' selected. The 'Report Name' is 'Daily Unbilled Report'. The 'Report Pswrd' field is empty. The 'Send as PDF' checkbox is checked. The 'Scheduled Time' is set to 7:00:00 AM. The 'Run' and 'Exit' buttons are at the bottom right.

Figure 2: The Automate Report tab, here on the Unbilled Report

14.1.1 Who receives it

In **Automated Report Users**, press the green + and pick a user from the **RepUser** drop-down. Their e-mail address fills in beside them. Add as many recipients as you need; the red – removes the selected one.

The address comes from the user's Mp10 record. If it is wrong or missing, fix it on the user, not here.

14.1.2 When it runs

Tick one of the **Frequency** options:

Option	Runs
Daily	Every day
Weekly	Once a week, on the day you select on the right
First day of month	On the 1st

Option	Runs
Last day of month	On the last day of the month
End of Each Period	On your billing period's closing day
Suspend	Not at all — see below

Scheduled Date runs the report once, on that specific date, instead of repeating.

End of Each Period uses the day-of-month set in BILLING / DAYS_INTO_MONTH_BEFORE_CLOSING.

14.1.3 What time it runs

Scheduled Time sets the time of day. If you leave it blank the system default is used:

Setting	Section	Default
AUTO_REPORTS_TIME	AUTOTASKS	22:00:00

A report is only queued for a time that has not yet arrived today. If it is 3 p.m. and you schedule a daily report for 2 p.m., it will not run this afternoon — it starts tomorrow. When setting up a report you want to see today, pick a time that is still ahead of you, or use **Run** to produce it immediately.

14.1.4 What format it arrives in

Tick one or more of **Send as PDF**, **Send as XLS**, **Send as Fp3**, **Send as HTML**. PDF is the usual choice. XLS is worth ticking when the recipient will want to sort or total the figures themselves.

Report Pswrd puts a password on the attachment. Use it for anything leaving the practice.

14.1.5 Naming and saving

Give it a clear **Report Name** — this is what identifies the scheduled job and what recipients will see. **Daily Unbilled Report** is a better name than **report1**.

Press **Save Automated Report**. Press **Run** to test it right away without waiting for the schedule.

14.2 Stopping a report

Tick **Suspend** and save. The job stays configured, with its recipients and settings, but stops running. Untick it to resume.

This is preferable to deleting a report you may want back.

14.3 Making sure e-mail actually goes out

Scheduled reports are e-mailed through the practice's outgoing mail server. If that is not configured, reports will run and produce nothing anybody sees.

These live in section **eMail** (*IT task*):

Entry	Example	What it is
SMTP-ServerName	smtp.gmail.com	Outgoing mail server
SMTP-Port	465	Its port
SMTP-User	reports@yourpractice.com	Account used to send
SMTP-Pass	(password)	Its password

Entry	Example	What it is
Address	reports@yourpractice.com	Address messages come from
SMTP-Auth	YES	Whether the server requires a login
SENDMETHOD	CDO or SMTP	How mail is sent

If you use Gmail or Microsoft 365, an ordinary account password will not work. You need an app-specific password, and the account must permit it. This is the most common reason scheduled reports stop arriving after an otherwise unrelated IT change.

14.4 Keeping a copy on disk

To have every automated report also written to a folder — useful as an audit trail, or to feed a shared drive:

Section	Entry	Example
AUTOREPORTS	COPY_TO_PATH	\\fileserver\mp10\reports\

Leave it blank to skip.

15 Modality worklist (imaging)

If your practice has imaging equipment — a CT, an ultrasound, an X-ray room — the **modality worklist** is how each machine finds out who its next patient is. Instead of a technologist typing the name, date of birth and accession number at the console, the machine asks Mp10 and fills its own screen in.

That matters for more than convenience. A study labelled by hand is a study that can be labelled wrongly, and a mistyped accession number is what makes an image arrive in the wrong patient's file.

This chapter covers the AutoTasks half only. AutoTasks decides which studies belong on the worklist and writes one small file per study. A **DICOM worklist server** — a separate piece, installed once on your server — is what the machines actually query, and it serves those files. Without it, AutoTasks is writing files nothing reads.

If you are setting this up for the first time, read the **Mp10 Modality Worklist** guide instead: it covers both halves, installing the server, pointing a scanner at it, and what to do when the worklist looks empty. This chapter is the AutoTasks settings reference that guide refers back to.

15.1 Turning it on — two separate switches

There are two, in this order, and the order matters:

```
Admin10 -> Global System Registry
|
| (1) GENERAL / MODALITY-WORKLIST = YES
v
run a file check in Admin10
|                                -> the modalityw1 table is created
v
| (2) AUTOTASKS / EXPORT_MWL = YES
v
restart AutoTasks
|
|                                -> the table is filled, and kept current
|                                on every cycle from then on
```

15.1.1 Why the settings are already there, set to NO

The first time Mp10 looks for a setting and does not find it, **it creates the entry itself**, set to its default, with a description of what it does. So the first time you go looking for MODALITY-WORKLIST you will probably find it already present and set to NO.

That is not a fault and not somebody else's change — it is Mp10 writing down a question it just asked. Once the entry exists, **your value always wins**; Mp10 never overwrites a value you have set.

15.1.2 Step 1 — create the table

Section	Entry	Set to
GENERAL	MODALITY-WORKLIST	YES

Then **run a file check in Admin10**. The table is created during that check and at no other time.

Order matters. The file check reads this setting as it runs. Setting it to YES *after* a check has finished does nothing until the **next** check. If you have just changed it, run the check again.

15.1.3 Step 2 — turn the reconcile on

Section	Entry	Set to
AUTOTASKS	EXPORT_MWL	YES

That takes effect on the next cycle, a couple of minutes later; settings are re-read every cycle and nothing needs restarting.

15.2 The settings

Section	Entry	Example	Default	What it means
GENERAL	MODALITY-WORKLIST	YES	NO	Whether the <code>modalitywl</code> table exists at all. Read during a file check, not by AutoTasks
AUTOTASKS	EXPORT_MWL	YES	NO	The on/off switch for keeping the table current
AUTOTASKS	MWL_SPOOL_PATH	C:\Mp10Mwl\worklist(<i>blank</i>)		Where the worklist files are written. Must match the worklist server's own setting exactly. Blank means no files are written
AUTOTASKS	MWL_WINDOW_DAYS	1	1	How far either side of today to look, in days
AUTOTASKS	MWL_UID_ROOT	(<i>blank</i>)	(<i>blank</i>)	IT task. Leave blank unless your organisation owns a registered DICOM root

15.2.1 MODALITY-WORKLIST — whether the table exists

Leave this NO and the table is never created, nothing else in this chapter applies, and no harm is done. Practices with no imaging equipment should leave it alone.

Setting it back to NO later does **not** delete the table or its contents. It only stops future file checks from maintaining it.

15.2.2 EXPORT_MWL — whether the table is kept current

With this NO, the table sits exactly as it was last left. With it YES, AutoTasks re-checks the whole worklist against your schedule on every cycle — by default every two minutes.

15.2.3 MWL_SPOOL_PATH — where the worklist files go

The worklist server watches one directory. This tells AutoTasks to write into that same directory, and the two settings must name **exactly the same place**.

Leave it blank and the table is still kept current but no files are written — which is a reasonable way to run for a while before the imaging side exists, and is what the log line `no .wl written -- AUTOTASKS/MWL_SPOOL_PATH is empty` is telling you.

Wrong path, empty worklist, no error. If the two settings disagree, each half does exactly what it was told: AutoTasks writes files, the server serves the empty directory it was pointed at, and nothing anywhere reports a problem. The scanner simply shows nothing. `Test-Mwl.ps1` on the server compares the two in one line, and is the fastest way to rule this out.

AutoTasks owns that directory. It writes what belongs there and deletes everything else in it — that is how a cancelled study disappears from the machines. Do not keep anything else in it.

15.2.4 MWL_WINDOW_DAYS — how wide the window is

1 means yesterday, today and tomorrow. 0 means today only. 7 means a fortnight centred on today.

Widen this with care. Everything inside the window is a live worklist entry that a technologist has to look past to find today's patient. A window of 30 is not a longer list so much as a worse one. Change it only if orders are genuinely being scheduled that far ahead.

Narrowing it does not lose anything permanently — see *Cancelling is not deleting* below.

15.2.5 MWL_UID_ROOT — how studies are identified (*IT task*)

Every study needs an identifier that is unique **worldwide**, not just within your practice, because images may end up in systems shared with other organisations.

Left blank — the default, and the right answer for almost everyone — Mp10 mints one from a scheme designed for exactly this and requiring no registration.

Fill it in **only** if your organisation genuinely owns a registered DICOM root. Digits and dots only.

Do not invent a value here. An invented root is not merely untidy: it can collide with identifiers issued by whoever really owns it, and two different studies sharing an identifier is a problem that surfaces years later, in somebody else's archive.

Changing this later is safe. Identifiers already issued are **never rewritten** — only newly-scheduled studies use the new root — so an existing study is never split in two.

15.3 Which orders appear on the worklist

An order is on the worklist when **all** of these are true:

- its encounter is **not** inactive,
- its encounter is **not** closed, and
- its start date falls within `MWL_WINDOW_DAYS` days either side of today.

Where an order has no start date of its own, the encounter's admission date is used instead.

There is nothing to switch on per order and nothing to tick. An order that meets those three tests is on the worklist; one that stops meeting them comes off it.

15.4 What happens on each cycle

AutoTasks works out the list from scratch every time, rather than trying to remember what changed. That is deliberate: it means a missed cycle, a service restart or a crash cannot leave the worklist permanently out of step. The next cycle simply puts it right.

Each pass does four things:

1. **Works out which orders belong** on the worklist right now.
2. **Adds** the ones that are not there yet, and **puts back** any that had been cancelled but now qualify again.
3. **Cancels** anything on the worklist that no longer qualifies.
4. **Writes a file for every study a machine should see**, and **deletes every other file** in the worklist directory. That last sweep is what actually withdraws a cancelled study from the machines.

Step 4 rewrites a file only when its content has changed — or when the file has gone missing, which is how a directory somebody has cleaned out repairs itself on the next cycle.

15.5 Cancelling is not deleting

When an order falls off the worklist — the encounter is closed, or the date moves out of the window — its entry is **marked cancelled, not removed**.

This matters if it comes back. Re-open the encounter, or move the date back into the window, and the *same* entry is reinstated with the *same* study identifier. The study is never split into two half-studies in your image archive.

The one case that does remove an entry outright is deleting the order itself, which is as it should be — an order that no longer exists cannot be scanned.

15.6 Telling the machine what kind of study it is

A worklist entry has to say whether the study is a CT, an ultrasound, an X-ray and so on. Mp10 works that out from the **revenue code** on the order: each revenue code carries the kind of imaging it represents.

Until those are filled in, entries are still created correctly in every other respect — they simply do not say what kind of study they are, which is not enough for a machine to act on. **This is a one-off data job**, done once per practice, and AutoTasks tells you how much of it is left (below).

15.7 What the log tells you

Every cycle writes a handful of lines to `AutoTasks.out`. Search it for `WriteWorklistFiles`:

Line	What it means
<code>active set size</code>	How many orders qualify for the worklist right now
<code>inserted / re-activated / unchanged</code>	What this cycle changed. On a quiet cycle all three are 0 and that is correct
<code>cancelled</code>	How many entries came off the worklist this cycle
<code>rows with a rev_code but no revcodes.Modality mapped</code>	How many entries cannot say what kind of study they are — with example revenue codes, so you know which ones to fill in. This number should fall to zero as the mapping is completed
<code>rows with no items/RevCode match at all</code>	Orders whose item does not resolve to a revenue code at all. Usually a catalogue problem rather than a worklist one

Then search it for `EmitWorklistFiles` — that is the part the machines see:

Line	What it means
written	Files written this cycle. High on the first run of the day, then usually 0
unchanged	Studies already published and unaltered. This is the number that should match what a scanner sees
purged	Files removed — cancelled studies, studies past the window, and anything else that was in the directory
no modality (not emitted)	Studies held back because their revenue code does not say what kind of study they are. These are invisible to every machine. Non-imaging orders — laboratory, office visits — are counted here too and belong here
write failures	Should always be 0. Anything else is a permissions or disk problem on the worklist directory
published with NO station	How many studies carry no imaging station. They answer a modality-and-date query but are invisible to a scanner filtering by station AE title. The line names example location/modality pairs — those are the missing Imaging Stations rows
station routing is OFF	The imaging stations table is absent or unreadable. The worklist is still published; nothing is routed
no .wl written -- AUTOTASKS/MWL_SPOOL_PATH is empty	The path is not set, so nothing is published
ABORTED -- AUTOTASKS/MWL_SPOOL_PATH does not exist	The path is set but wrong, or the directory was deleted

A healthy log on a normal day shows a steady `active set size`, small or zero change counts, a zero mapping count, and `written 0 / unchanged <today's studies>` on most cycles.

15.8 Running one cycle by hand

You do not have to wait two minutes and read around everything else the service does. On the server:

```
AutoTasks.exe --mwl-run
```

That runs **one worklist cycle and nothing else** — no backup, no remittances, no e-mail, no texts — and prints the same lines to the screen. It is safe to run in the middle of a working day, and it is the quickest answer to “is the worklist right *now*”.

16 Troubleshooting

Start with `AutoTasks.out`. Search it for the date you expected something to happen. Almost every question below is answered by a line in that file.

16.1 Nothing at all is happening

Check	How
Is the service running?	Windows Services , look for the Mp10 AutoTasks service (<i>IT task</i>)
Is it writing to the log?	Check the timestamp on <code>AutoTasks.out</code> — it should update every couple of minutes
Were settings changed less than a cycle ago?	Wait one cycle — they are re-read every time, no restart needed. If a change has still not taken effect after that, check the spelling of the entry: a misspelt one is a setting nobody reads

16.2 The backup never runs

Symptom	Likely cause
Nothing in the log about a backup	One of <code>BACKUP_PATH</code> , <code>BACKUP_TIMES</code> , <code>BACKUP_DAYS</code> is blank — all three are required
“Backup started” then an error	The path is not reachable <i>by the database server</i> . Check for a mapped drive letter; use UNC instead
Runs some days but not others	Check the digits in <code>BACKUP_DAYS</code> against the day-code table — 1 is Sunday, not Monday
Ran once, never again the same day	Working as designed — one backup per calendar day
Zips are disappearing	<code>MAX_BACKUPS_TO_KEEP</code> is pruning them. Raise it or move zips off elsewhere

16.3 No backup e-mail arrived

Check the log first — it distinguishes “the backup did not run” from “it ran but the e-mail did not go out”, and those need different fixes.

In <code>AutoTasks.out</code>	Meaning
Backup notice sent to N recipient(s)	It was sent. Look in the recipients’ spam folder
Backup notice FAILED to send to	The mail server rejected it — check the eMail settings
Backup notice could not be sent (mail error)	The mail server could not be reached at all
notifications are switched off	<code>NOTIFY_EMAIL</code> is blank
holds no usable address	Nothing in <code>NOTIFY_EMAIL</code> looks like an address — check for a typo
Nothing at all about a notice	The backup itself never completed. See <i>The backup never runs</i>

Symptom	Likely cause
Only one message instead of two	The second comes from the zip step. If <code>BACKUP_ZIP_PATH</code> is blank there is nothing to archive, and the first message says so
Some recipients got it, others did not	An entry in the list is not a valid address and was skipped. The log line gives the count actually used
A FAILED message, then nothing more that day	Working as designed — AutoTasks does not retry a failed backup until the next day
No message at all, on a night you expected one	The service is not running, or never reached the backup. Silence is not confirmation — check the log
No outgoing mail server is configured in the log	The eMail settings are blank. The backup still ran; only the notice was skipped
already sent today; not repeating in the log	A notice of that kind had already gone out today

16.4 Remittance PDFs are not being picked up

Symptom	Likely cause
Files sit in the folder, nothing in the log	<code>SCANNING / EOBIMAGESPATH</code> is blank or misspelled, or the service was not restarted
“belongs to another practice - left in place”	Working as designed — the cheque is payable to a different practice on this server
The file was read but no payments appeared	The claims on it could not be matched. The log names each claim it could not resolve
Log mentions the extractor failing	Look in <code>python_debug.log</code> . Usually Python or the extractor script is missing from the server
Only scanned remittances fail	Poppler or Tesseract is not installed

Once processed, the PDF is in the `history` sub-folder — it has not been lost.

16.5 835 files are not being processed

Symptom	Likely cause
The same remittances are processed over and over	<code>X12FILE_PATH / 835_COPY_PATH</code> is blank, or the history folder does not exist, or the service cannot write to it. The copy fails, so the erase is skipped. This is the most common 835 problem
Files sit in the folder, nothing in the log	<code>AUTOTASKS / PROCESS_835</code> is not YES , or the service was not restarted after changing it
Some files are processed, others ignored entirely	The ignored ones do not have 835 in the file name. Only <code>*835*</code> is looked at
File more than 6 mo old in the log	Working as designed — anything over 180 days is skipped
<code>X12 835 dir ... does not exist</code> at startup	<code>X12FILE_PATH / 835</code> is wrong, or is a Windows path on a Linux server
<code>is not a legit 835</code> in the log	The file is not a valid 835 — often an acknowledgement or a report the clearinghouse routed alongside the remittances

Symptom	Likely cause
Nothing is ever downloaded	<code>AUTOTASKS / GetRoutedFiles</code> is not <code>YES</code> , today is not in <code>GetRoutedFilesDays</code> , or the time is outside the start/end window
Error connecting to Host in the log	The <code>X12WebService</code> user name or password is wrong, or the server cannot reach the clearinghouse

16.6 Nothing is being downloaded from the clearinghouse

Symptom	Likely cause
0 routed files downloaded, every time	Normal if the clearinghouse has nothing new. If it never changes, check <code>AUTOTASKS / GetRoutedFiles</code> is <code>YES</code>
Nothing after 7 p.m.	Working as designed — there is a fixed cut-off no setting can move
Nothing on certain days	Today is not in <code>GetRoutedFilesDays</code>
Error connecting to Host	<code>X12WebService</code> user name or password, or the server cannot reach the clearinghouse
Files download but 277s are never processed	<code>X12FILE_PATH / 277</code> points somewhere other than <code>X12FILE_PATH / 835</code> . Everything is downloaded into the 835 folder
A file was collected once and never again	It was marked as downloaded at the clearinghouse. Retrieve it from the portal by hand

16.7 Claim statuses or acknowledgements are not appearing

Symptom	Likely cause
Nothing happens at all	<code>AUTOTASKS / PROCESS_277</code> or <code>ProcessedIDs</code> is not <code>YES</code>
The same files process repeatedly	<code>X12FILE_PATH / 277_COPY_PATH</code> is blank, missing, or not writable
Unlegit 277 in the log	Normal — that is a file in the shared in-tray that is not a 277, most often an 835
Acknowledgements vanish without reaching history	Working as designed — <code>TA1_</code> and <code>999_</code> files are deleted, not filed

16.8 Eligibility is not being checked

Symptom	Likely cause
No patients at all are checked	<code>AUTOTASKS / AutoSend_270</code> has been set to something other than <code>YES</code>
One insurer works, another does not	The failing insurer's plan record: <code>isFetch271</code> not ticked, or <code>Submitter / Receiver / GS03</code> blank
An established patient is never checked	Their insurance record has not changed in the last 7 days. Re-save it to bring it back into the window
A secondary insurance is ignored	Only the primary is checked

16.9 The modality worklist is empty or wrong

Symptom	Likely cause
The <code>modalitywl</code> table was never created	GENERAL / MODALITY-WORKLIST was set to YES <i>after</i> the last file check, not before. Run the file check again
The table exists but stays empty	AUTOTASKS / EXPORT_MWL is not YES , or the service was not restarted after it was set
<code>active set size</code> is 0 but there are orders today	Their encounters are closed or inactive — both are excluded — or the order dates fall outside MWL_WINDOW_DAYS
Everything is suddenly cancelled	MWL_WINDOW_DAYS was narrowed, or a batch of encounters was closed. Nothing is lost; widen it or re-open them and the same entries come back
Entries exist but say nothing about the kind of study	The revenue-code mapping is not filled in. The log line counting these names example revenue codes to start with
The table looks right but no files are written	AUTOTASKS / MWL_SPOOL_PATH is blank or names a directory that does not exist. The EmitWorklistFiles log line says which
Files are written but a scanner sees nothing	The two halves are pointed at different directories, or the DICOM server is not running. Run Test-Mwl.ps1 on the server; the Mp10 Modality Worklist guide covers this in full
A study is on the table but has no file	It has no modality — see no modality (not emitted) in the log. A laboratory or office-visit order belongs in that count

The file check does not complain when a table cannot be created. If the `modalitywl` table is missing after a check that appeared to finish normally, do not assume the setting is wrong — the check can fail silently on a structure problem. Report it rather than re-running it repeatedly. (*IT task.*)

16.10 Appointment reminders are not going out

Symptom	Likely cause
No texts at all	SMS / SEND_APPOINTMENT_REMINDERS is not YES , or APPOINTMENT_REMINDERS_ALERT_DAYS is 0
Queued but never sent	The TWILIO account settings — check ACCOUNT_SID , AUTH_TOKEN and TEL_FROM
Texts stop partway through the day	MAX-DAILY-SMS reached. The log says so explicitly
A patient with two appointments got one text	Working as designed — one reminder per patient per day
An appointment just entered was skipped	It is inside the COOLING_PERIOD — it is texted on a later cycle
Nothing for a specific patient	No mobile number on the appointment, a number under 10 digits, no time on the appointment, or it is already confirmed

16.11 A scheduled report did not arrive

Symptom	Likely cause
It never ran at all	Check Suspend is not ticked; check the frequency matches today
Set up today, did not run today	The scheduled time had already passed when you saved it — it starts tomorrow
Ran, but nobody received it	The eMail settings, most often an expired or non-app-specific password
One person did not get it	Their address on their user record, not on the report
Arrived empty	The report's own parameters — run it manually with Run and see

16.12 Getting help

When contacting support, having these ready will save a round trip:

- the relevant section of **AutoTasks.out**, covering the time in question
- **python_debug.log**, for remittance PDF problems
- **backup_zip.log** and **backup_zip.stdout.log**, for backup archive problems
- the exact Section and Entry values you have set
- for a remittance problem, the PDF or the 835 file itself

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